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EFFECTS OF HOLE SHAPE ON FILM COOLING WITH LARGE ANGLE INJECTION

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ABSTRACT

In this study the film cooling performance of a single row of discrete holes inclined at an injection angle of 55° is investigated at a density ratio of $DR = 1.6$. Three different hole geometries were used in this study, a round hole and two shaped holes. One shaped hole had forward and lateral expansions of 15°, and the other a 15° lateral with a 25° forward expansion. For reference, a round hole with an injection angle of 35° was also tested. The film cooling performance of each hole shape was evaluated using adiabatic effectiveness, thermal field, and velocity field measurements.

The shaped holes showed higher spatially averaged adiabatic effectiveness than the round hole over the whole range of momentum flux ratios (I) investigated. The effectiveness values for the shaped holes were only marginally better than the round hole at the low I, but at the high I, the shaped holes performed much better than the round hole. The temperature and velocity field measurements near the hole exit suggest that there is a slight detachment of the jet from the wall for the round hole, while the jets remain attached for the two shaped holes. The shaped hole with the larger forward expansion had a warmer jet with a higher trajectory at the hole exit suggesting ingestion of mainstream fluid and flow separation within the hole.

INTRODUCTION

Film cooling in gas turbine airfoils is usually achieved through discrete holes inclined at shallow angles to the blade surface. However, in some cases structural constraints favor the use coolant holes with larger injection angles. Most studies in the film cooling literature have used injection angles of 30° to 35°; however, film cooling with an injection angle of 55° was studied by Foster and Lampard (1980) and, more recently, by Kohli and Bogard (1997). Not surprisingly, the performance of holes inclined at 55° was poorer than for the 35° holes, but maximum adiabatic effectiveness was actually similar for both hole inclinations. The main detriment for the 55° holes was a more rapid decrease in adiabatic effectiveness as blowing ratios increased. This was due to detachment of the coolant jet at lower blowing ratios for the higher angle injection. In this study of film cooling with large angle injection, we investigated the improvement of performance made possible by shaping the hole exit with lateral and forward expansion angles.

There have been relatively few studies of the effect of shaping film cooling holes reported in the literature, and these have all used an injection angle of 35°. The effects of hole shape on film cooling performance were first reported by Goldstein, Eckert, and Burggraf (1974). This study compared the effectiveness for shaped holes, which had a 10° lateral expansion near the exit, to that of round holes. The centerline adiabatic effectiveness of the round and shaped holes were comparable for low blowing ratios, but at high blowing ratios the centerline adiabatic effectiveness of the shaped holes was as much as a factor of three higher than that for the round holes. According to Goldstein et al., this improvement in film cooling effectiveness for the shaped holes was due to the increase in exit area which reduced the momentum flux of the jet, and thereby decreased penetration into the freestream. A significant limitation in this study was most of the comparison were based on centerline effectiveness, and improvements to laterally averaged adiabatic effectiveness due to improved lateral spreading of the film cooling jets was not addressed.

Film cooling using shaped holes was also investigated by Makki and Jakubowski (1986) who compared the performance of diffused trapezoidal shaped holes with round holes with an injection angle of 35°. The performance of the film cooling holes was evaluated in terms of the reduction of the heat flux rate from the surface. The reductions in heat flux for the shaped holes were significantly greater than that for round holes for most blowing conditions. However, for the high blowing ratios used in this study, coolant jets from the round holes

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would be expected to be fully separated from the surface and hence result in poor performance. More recent studies by Thole et al. (1998) and Gritsch et al. (1998) compared the performance of a round film cooling hole with two shaped holes. These holes were $6D$ in length and inclined at 35°. Two expansion shapes were investigated, the first had a lateral expansion of 14° along a $4D$ length, and the second combined the 14° lateral expansion with 15° forward expansion along a $1D$ length. Thole et al. studied the velocity fields at the exit of these holes and downstream, and Gritsch et al. studied the adiabatic effectiveness. Because a single hole was used in these studies, rather than a row of holes, laterally averaged adiabatic effectiveness was estimated by integrating the local effectiveness over a spanwise range of 5.5 diameters. The greatest difference in performance between the round holes and shaped holes was a factor of three increase in laterally averaged effectiveness for the largest blowing ratio studied, $M = 1.5$. The increase in adiabatic effectiveness was attributable to two major factors: the wider lateral distribution of coolant from the shaped holes, and the coolant jets from the shaped holes appeared to be less susceptible to detaching from the surface at high blowing ratios.

A unique aspect of this study is the determination of the improvement of performance possible using hole exit shaping with film cooling injection at large angles. Laterally averaged, and spatially averaged effectiveness were measured over a range of blowing conditions using rows of holes with injection angles of 55°. Three different hole geometries were used in this study, a round hole and two shaped holes. One shaped hole geometry had forward and lateral expansions of 15°, hereafter referred to as shape A. The other, referred to as shape B, had a 15° lateral and a 25° forward expansion. Also, as a standard for comparison, effectiveness measurements were done for round holes inclined at 35°. Furthermore, details of the coolant jet trajectories and dispersion were obtained from thermal field measurements in the hole, at the hole exit, and downstream of the hole. Velocity field and turbulence measurements were also made at the hole exit and downstream of the hole.

EXPERIMENTAL FACILITIES AND TECHNIQUES

Experiments were carried out in a closed-loop wind tunnel with a turbulence level of $TL = 0.2\%$ and a secondary cryogenic injection system. To obtain a density ratio of $DR = 1.6$, liquid nitrogen was used to cool the injected air to about -85 °C (with $T_\infty \sim 25$ °C). Details of the facility can be found in Pietrzyk, Bogard and Crawford (1990). For the experiments in this study, the freestream velocity was varied over the range $U_\infty = 10$ to 30 m/s. The approaching boundary layer for a freestream velocity of $U_\infty = 20$ m/s was turbulent, and the boundary layer parameters were $Re_\theta = 1100$, $\delta^*/D = 0.12$, and $H = 1.48$. A detailed documentation of the freestream and turbulent boundary layer quality can be found in Pietrzyk et al. (1990).

A test plate identical to that described in Kohli and Bogard (1997) was used for this study. To ensure adiabatic conditions this test plate was constructed of very low conductivity polystyrene. For this study a single row of nine holes inclined at 55° was used for each hole geometry. Figure 1 shows details of the three hole geometries used in this study: a round hole, a shaped hole with 15° lateral and forward expansion, and, another shaped hole with a 15° lateral and 25° forward expansion. The same metering length, $L_M/D = 0.9$ was used for the two shaped holes. All holes had a diameter of $D = 11.1$ mm, were spaced $P/D = 3$ apart and had a length, $L/D = 2.8$, representative of actual gas turbines. The hole inlets and exits were sharp edged, and the interiors were aerodynamically smooth.

Wall temperature measurements were made with ribbon-type thermocouples glued to the surface of the adiabatic plate on the centerline of the central hole at $x/D = 2, 3, 5, 6, 8, 10, 15, 22, 30$ and 50. Additional off-center thermocouples were added to form four spanwise arrays of thermocouples located at $x/D = 3, 6, 10$ and 15. Each spanwise array consisted of seven thermocouples equally spaced between $-1.5 \le z/D \le 1.5$, giving the surface temperature (adiabatic wall temperature, T_{aw}) distribution across one hole pitch. The laterally averaged effectivenesses determined at these four stations were used to obtain spatially averaged effectiveness ($\bar{\bar{\eta}}$) over $3 \le x/D \le 15$. These effectiveness measurements were done over a range of blowing conditions in order to identify the range producing the maximum adiabatic effectiveness for each hole shape.

Detailed thermal field measurements were made at one blowing ratio which was in the optimal operating range for each hole shape using a 2 mil thermocouple probe. In order to obtain the three-dimensional thermal field for the jet, lateral planes of temperature data

NOMENCLATURE

AR = area ratio for shaped holes, see Figure 1
D = diameter of film cooling hole
DR = density ratio of jet to freestream = ρ_j / ρ_∞
H = shape factor = δ^*/θ
I = momentum flux ratio of jet to freestream = $\rho_j U_j^2 / \rho_\infty U_\infty^2$
L = hole length
L_M = metering length of film cooling hole
M = mass flux (blowing) ratio of jet to freestream = $\rho_j U_j / \rho_\infty U_\infty$
P = hole spacing
Re_θ = Reynolds number = $U_\infty \theta / \nu$
T = temperature
TL = turbulence level = $[(u_{rms} + v_{rms}) / 2]^{1/2} / U_\infty$
U = mean velocity, x direction
u = fluctuating velocity, x direction
V = mean velocity, y direction
v = fluctuating velocity, y direction
x = streamwise distance measured from downstream edge of hole
y = wall normal distance
z = spanwise distance measured from centerline of hole
δ^* = displacement thickness
θ = momentum thickness
η = adiabatic wall effectiveness = $(T_{aw} - T_\infty)/(T_j - T_\infty)$
ρ = density
ν = viscosity
Θ = normalized temperature = $(T - T_\infty)/(T_j - T_\infty)$

Subscripts

aw = adiabatic wall
cl = centerline
j = jet
rms = root mean square
∞ = freestream

Superscripts

$\bar{}$ = lateral average
$\bar{\bar{}}$ = spatial average

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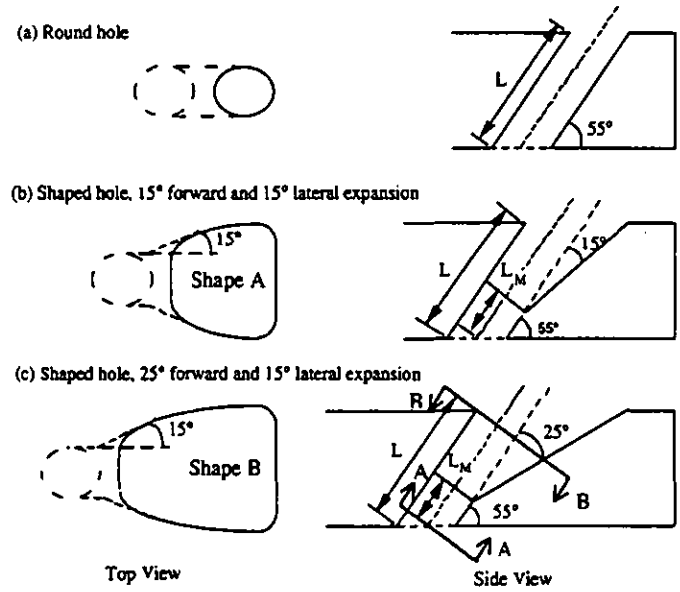
were recorded for all three geometries at $x/D = 0, 3, 6, 10, 15$ and 30. Temperature profiles were also measured on the centerline downstream of the hole at $x/D = 0.5, 1, 2, 5, 8, 22$, and 50. In addition, profiles were measured at three or four different positions over the hole and along the jet centerline; the positions of these measurements varied depending on the hole geometry. For all thermal field measurements, the temperature at each measurement position was obtained by using a sample size of 100 points over a duration of 100 msec. This sampling was shown to provide a precision accuracy for mean temperature measurements of $\Delta\Theta = \pm 0.01$.

For the experiments conducted in this study the mass flux ratio for each jet was held constant within $\delta M = \pm 5\%$. This includes variations in the overall secondary flow loop and variations between individual jet velocities. The variation in density ratio due to changes in the jet and freestream temperatures was estimated as $\delta DR = \pm 2.5\%$. Bias errors for the adiabatic effectiveness measurements due to conduction and radiation losses were estimated and found to be negligible. Consequently, the uncertainty of the adiabatic effectiveness measurements was primarily due to random variations of the test conditions. To determine the magnitude of this uncertainty, repeatability tests were conducted at $I = 0.1$ for the round hole in this study. Statistical analysis of the variation in local, laterally averaged, and spatially averaged effectiveness obtained from multiple data sets and different runs, indicated the following uncertainties: $\delta\eta = \pm 0.03$, $\delta\overline{\eta} = \pm 0.02$, and $\delta\overline{\overline{\eta}} = \pm 0.02$. The uncertainty in thermal field results at a nominal value of $\overline{\Theta} = 0.5$ due to variation in the freestream and jet temperatures, and the uncertainty in the temperature measurement was $\delta\overline{\Theta} = \pm 2.5\%$.

RESULTS AND DISCUSSION

Adiabatic effectiveness performances for the different hole geometries are compared at blowing conditions of $I = 0.16$ ($M = 0.5$) and $I = 0.63$ ($M = 1$) in Figure 2 and Figure 3, repectively. These represent a moderate momentum flux ratio and a high momentum flux ratio for round hole injection. In each case, results for the centerline and laterally averaged effectiveness are presented. Note that mass and momentum flux ratios were defined in terms of the average velocity at the inlet of the hole, so equivalent mass flow rates are obtained from different holes geometries when M is matched. At $I = 0.16$ the round holes with 55° injection are at optimum performance, i.e. producing maximum adiabatic effectiveness. Centerline adiabatic effectivenesses were fairly similar for each hole geometry, and this may be attributed to the coolant jets for all four hole geometries essentially remaining attached. However, in terms of laterally averaged adiabatic effectiveness the shape A holes have distinctly better performance near the holes, even relative to the round holes with 35° injection. This would indicate that shape A holes produce a greater lateral distribution of coolant. The performance of the shape B holes, although better than the round holes with 55° injection, was less than that for the shape A holes. This decrease in performance for the shape B holes is discussed later in this paper. At the higher momentum flux ratio of $I = 0.63$, the shaped holes have distinctly better performance, especially in terms of the laterally averaged adiabatic effectiveness. This is primarily due to the decrease in performance for the round holes because the coolant jets for these holes have detached from the surface. At $x/D = 3$, η_{CI} and $\overline{\eta}$ for the shape A holes decrease at $I = 0.6$ relative to $I = 0.16$. This decrease in adiabatic effectiveness with increase in mass flux



$L/D = 2.8$; $L_M/D = 0.9$; AR = area at exit plane BB / area at inlet plane AA

Figure 1. Details of the film cooling geometries used in this study.

ratio indicates some detachment of the coolant jet from the surface for the shaped holes near the holes. However, by $x/D = 15$, there is a 20% increase in adiabatic effectiveness for the shape A holes at higher blowing ratio. This suggests that the coolant jets have reattached to the surface at this greater distance downstream from the holes.

Although η_{CI} also decreases for the 35° round holes, the smaller decrease suggests that the jet detaches and re-attaches to the surface. This is consistent with the results of Thole, Sinha, Bogard and Crawford (1992) who found that for $0.4 \bullet I \bullet 0.8$, the film cooling jet for a 35° round hole re-attaches to the wall after lifting off.

Performances of the different hole geometries as a function of the momentum flux ratio were compared by calculating the spatially averaged effectiveness, $\overline{\eta}$, over the range $3 \bullet x/D \bullet 15$. These results are presented in Figure 4. Immediately apparent from this figure is that the performance of the round holes and the shaped holes are similar for $I < 0.1$ ($M = 0.4$), but the round holes reach a peak performance shortly after this, while the adiabatic effectiveness for shaped holes continues to increase reaching a peak at $I = 0.35$ ($M = 0.75$). The shaped holes continue to increase adiabatic effectiveness as the momentum flux ratio increases to $I = 0.35$ because the coolant jet remains attached to surface. Recognizing that adiabatic effectiveness will begin to decrease with increasing blowing ratio when decreases caused by the detachment of the coolant jet offset increases due to greater mass flux of coolant, the peak adiabatic effectiveness at $I = 0.35$ suggests that the coolant jet starts to significantly detach at this point. The rationale for using shape B was that the increased forward expansion relative to shape A would have the potential to further increase the maximum adiabatic effectiveness and would potentially extend the range of momentum flux ratio over which the coolant jets remains attached. However, as shown in Figure 4, the performance of shape B was actually slightly worse than shape A. An explanation for

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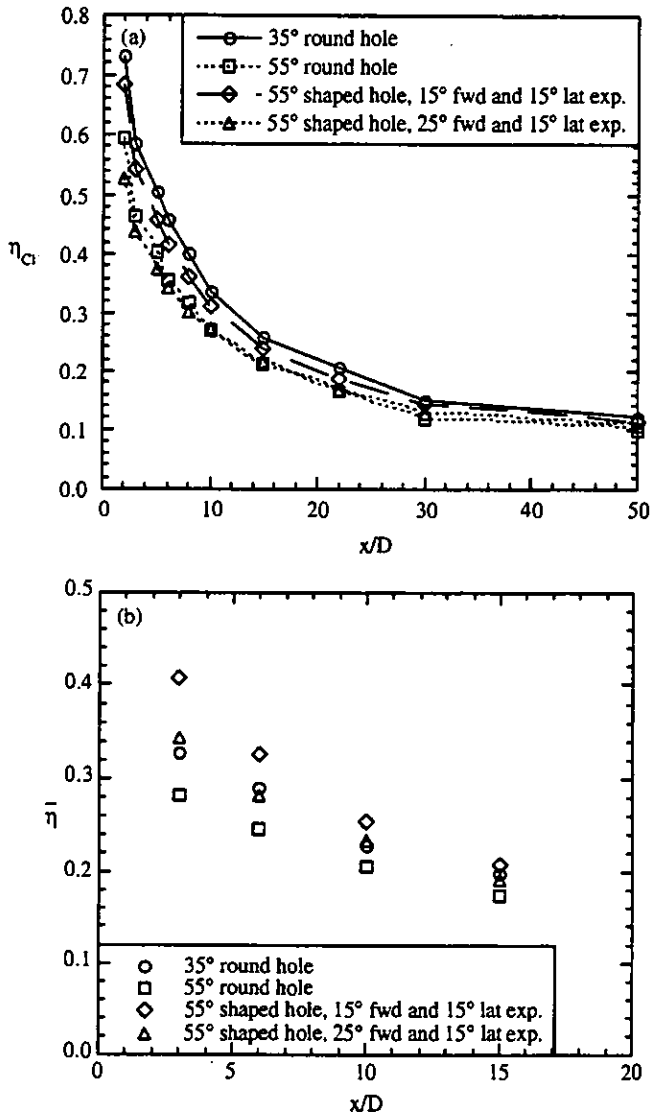

Figure 2. (a) Centerline and (b) laterally averaged effectiveness at $I = 0.16$, $M = 0.5$, $DR = 1.6$.

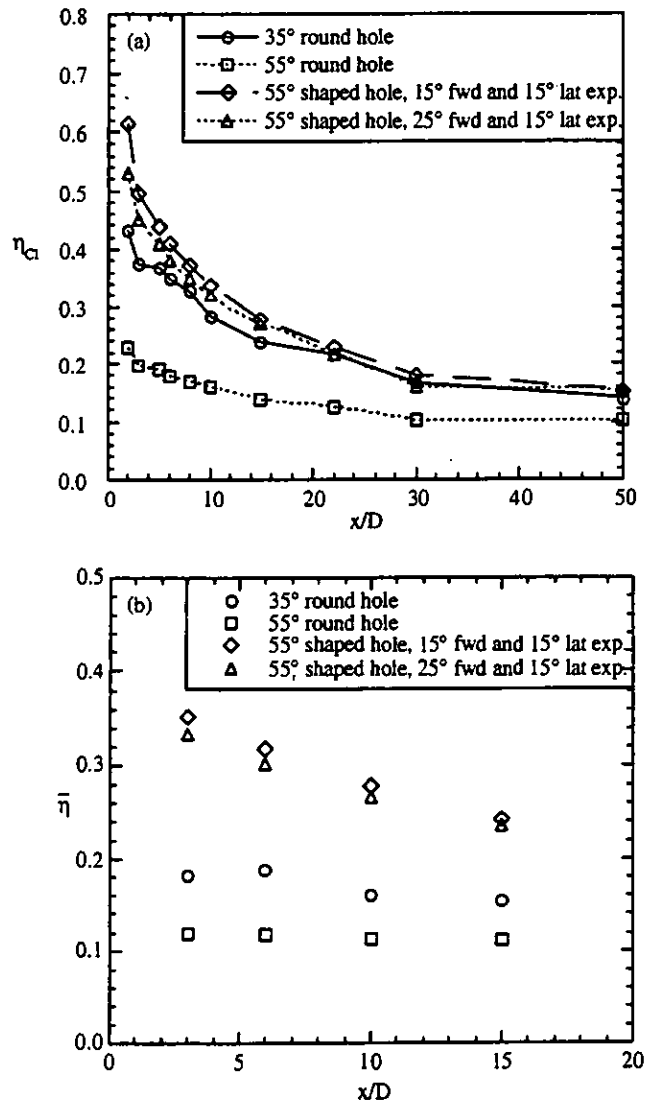

Figure 3. (a) Centerline and (b) laterally averaged effectiveness at $I = 0.63$, $M = 1.0$, $DR = 1.6$.

this decrease in performance is discussed later in this paper in conjunction with presentation of the thermal field measurements.

The increase in performance for the shaped holes might be attributed to a decrease in the exit velocity due to the diffusing expansion at the hole exit. Area ratios were calculated based on the the area normal to the axis of the hole at the exit and inlet of the hole as shown in Figure 1. For the shape A holes the area ratio for the expansion was $AR_A = 2.9$, so a perfect expansion would have resulted in a decrease in exit velocity by a factor of 2.9. Effective mass flux ratios and momentum flux ratios based on this exit velocity would decrease by factors of 2.9 and 8.2, respectively. However, as discussed previously, the momentum flux ratio for peak adiabatic effectiveness increased by a factor of three for the shaped holes relative to the round holes. Since this increase in the optimum momentum flux ratio was significantly less than would be predicted based on the exit velocity of a perfectly expanded diffuser, the flow through the shaped holes was probably not perfectly expanded.

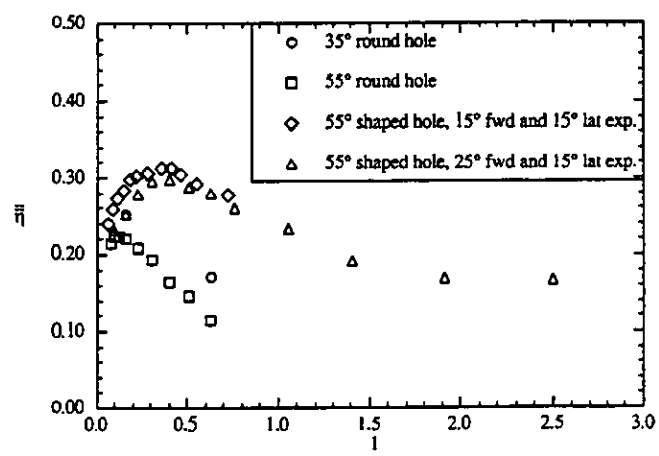

Figure 4. Comparison of spatially averaged effectiveness for all hole geometries.

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Although a full expansion did not occur in the shaped holes, the large increase in peak effectiveness for the shaped holes indicates at least a partial expansion of the coolant jet, resulting in a decrease in the exit velocity, and hence a higher mass flux ratio (based on the inlet area) before separating. These results suggest the possibility of modeling the shaped holes as round holes with equivalent diameters significantly larger than the actual inlet diameter. Consequently, the effective P/D would be smaller, and this would account for larger laterally averaged adiabatic effectiveness. However, measurements of the thermal field, presented below, show distinct differences in the coolant jets downstream of the round and shaped holes which preclude modeling the shaped holes as enlarged round holes.

Note that high $\bar{\bar{\eta}}$ values are achieved over a broad range of I for the shaped holes, suggesting that the performance of these holes is not very sensitive to the blowing condition. This characteristic is of great practical importance because precise control of blowing conditions in an actual turbine is not feasible. Therefore, small deviations from the optimum I would cause severe reduction in performance for the round holes, while the shaped holes would still maintain high levels of effectiveness.

While the effectiveness results clearly indicate that the shaped holes perform better than the round hole, thermal field measurements are essential for explaining why. Thermal field measurements were done at momentum flux ratios which gave maximum $\bar{\bar{\eta}}$ for each hole geometry. For the 55° round hole, the thermal field was mapped at $I = 0.1$ ($M = 0.4$). For the shaped holes, thermal field measurements

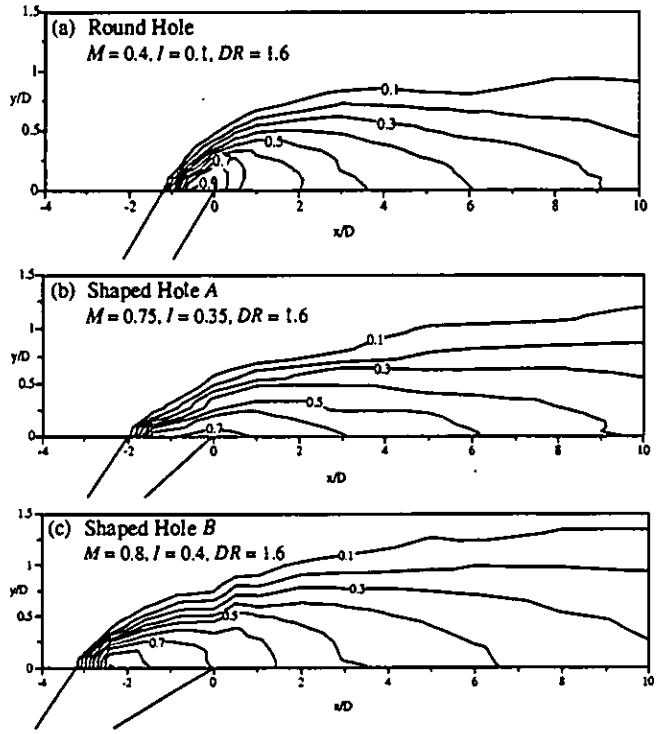

Figure 5. Centerline temperature contours ($\bar{\Theta}$) at optimum I for each hole geometry.

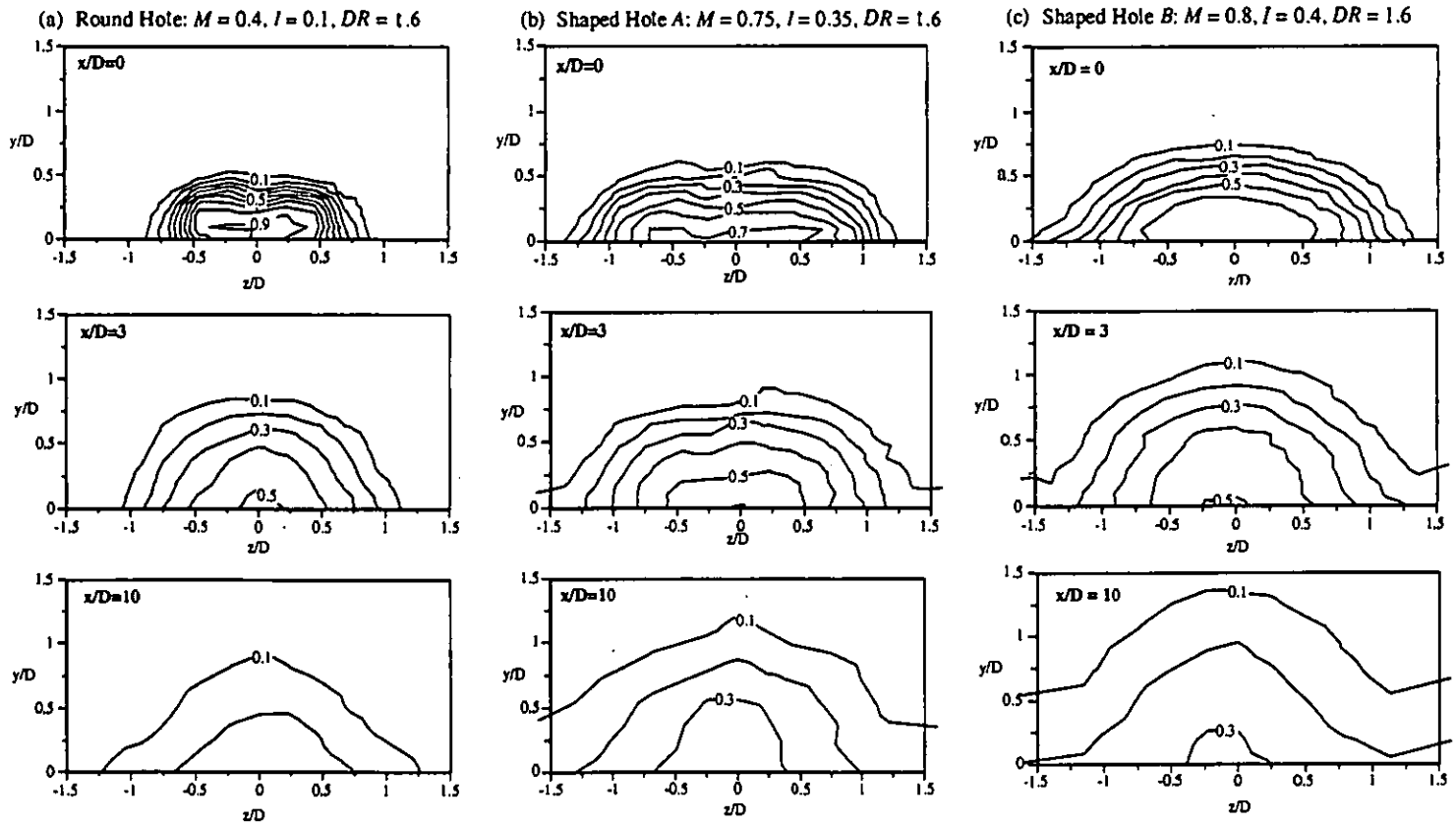

Figure 6. Lateral temperature contours ($\bar{\Theta}$) at optimum I for each hole geometry at various streamwise locations.

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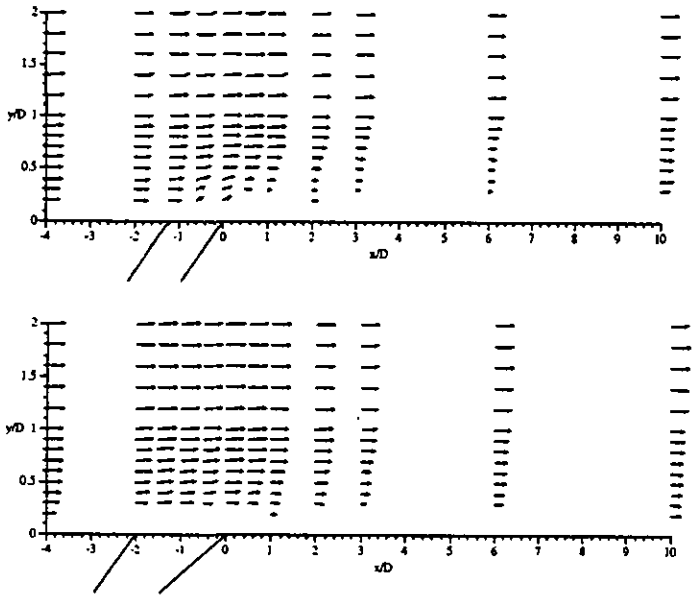

Figure 7. Mean velocity vectors along the centerline at the optimum I for each hole geometry; (a) round hole at $M = 0.4$, $I = 0.1$, $DR = 1.6$, (b) shaped hole A at $M = 0.75$, $I = 0.35$, $DR = 1.6$.

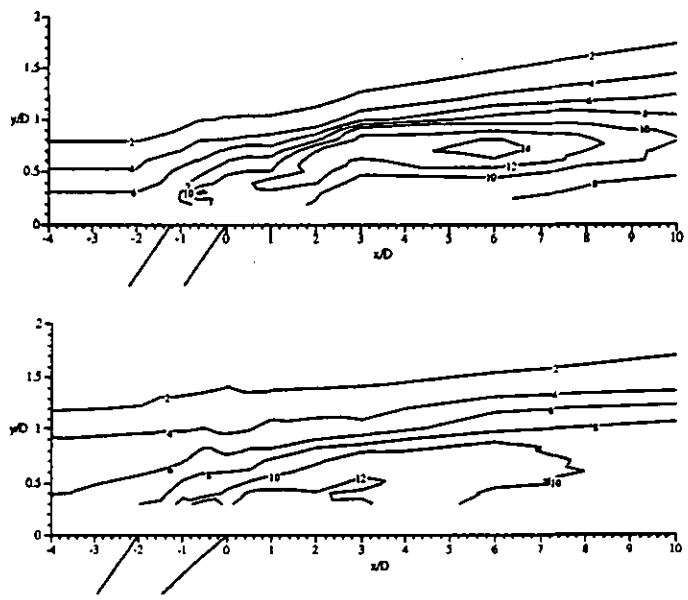

Figure 8. Turbulence level contours along the centerline at optimum I for each hole geometry; (a) round hole at $M = 0.4$, $I = 0.1$, $DR = 1.6$, (b) shaped hole A at $M = 0.75$, $I = 0.35$, $DR = 1.6$.

were made at $I = 0.35$ for shape A ($M = 0.75$), and $I = 0.4$ ($M = 0.8$) for shape B. Normalized temperature contours, $\bar{\Theta}$, of the film cooling jets along the centerline are shown in Figures 5(a) through (c). The penetration of the coolant jets into the mainstream, evident from the height of the $\bar{\Theta} = 0.1$ contour level, was similar for each of the hole geometries even though the shaped holes had a factor of 3.5 to 4 larger momentum flux ratio. The coolant jets remain attached for all hole geometries, but this would be expected since in each case the conditions were that which provided maximum adiabatic effectiveness.

There was a noticeable difference in the exit temperature of the coolant for the round and shaped holes. For the round holes normalized values were in excess of $\bar{\Theta} = 0.9$, while for the shaped holes they were slightly above $\bar{\Theta} = 0.7$ and $\bar{\Theta} = 0.8$ for shapes A and B, respectively. These results suggest that, for the shaped holes, there is some dilution of the coolant gas with mainstream air before exiting the hole, i.e. that there is an ingestion of mainstream fluid into the hole. This mixing of mainstream fluid with the coolant gas within the hole was confirmed with temperature measurements extending $1.5D$ into the holes. These measurements showed that within the shaped holes the coolant temperature increased significantly deep within the hole which can only be attributed to ingestion of mainstream fluid into the hole. A similar ingestion was predicted in the computational simulation reported by Kohli and Thole (1997) for an axially directed shaped hole with similar 15° expansion angles, but with a 35° injection angle.

Figure 5 also shows that shape A holes have maximum $\bar{\Theta}$ levels at the downstream exit of the holes, but shape B holes have maximum $\bar{\Theta}$ levels towards the upstream side of the holes. This result suggests that the shape B holes were not effective in turning the coolant jets. Thermal field measurements within the hole also indicated that separation occurred resulting in the bulk of the coolant flow continuing along the upstream edge of the hole. This also causes a higher trajectory for the coolant jet exiting the shape B holes, relative to the shape A holes, which is evident in the greater outward bulge in $\bar{\Theta}$ contours over the holes. These results explain the inferior performance for the shape B holes even though these holes expand to a greater exit area than the shape A holes.

The lateral distribution of the coolant distributed from each of the hole geometries is evident in the thermal field contours presented in Figure 6. At $x/D = 0$, coolant emerging from the round holes shows a well defined core with sharp temperature gradients surrounding this core. Coolant emerging from the shaped holes is at lower $\bar{\Theta}$ levels due to mixing with mainstream fluid within the hole, and this also causes broader temperature gradients around the coolant jets. Both shaped holes have distinctly broader distributions of coolant than the round holes, but the vertical extent is similar (shape B holes are slightly higher). Note that the difference in the coolant distribution between round and shaped holes is significant because it shows that these shaped holes can not modeled as round holes with larger effective exit areas. The shaping of the holes leads to a flattening of the coolant distribution which provides improved coverage and better attachment to the surface.

By $x/D = 3$ downstream of the holes, the coolant jets from the shaped holes have merged with adjacent jets, while the jets from the round holes still have not merged by $x/D = 10$. The better performance for the shape A holes can be attributed to the broader distribution of the coolant at each streamwise position.

Velocity field measurements were made along the centerline through the holes for the large angle round holes and the shape A holes. Blowing conditions for these measurements were the same as that for the previously described thermal field measurements, i.e. $I = 0.1$ ($M = 0.4$) for the 55° round hole, and $I = 0.35$ ($M = 0.75$), for

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shape *A* holes. Mean velocity field measurements are presented in Figure 7 in the form of *U-V* vectors. There was a distinct difference in the velocity field over the holes with the round hole showing a much greater *V* component of velocity compared to the shaped hole. This is a direct indication that shaping the hole effectively turns the coolant jet towards the streamwise direction even though the momentum flux ratio for the shaped hole is over three times greater than for the round hole. There was also a larger region of decelerated flow downstream of the hole for the round hole compared to the shaped hole. This is most evident in contrasting the velocity profile at $x/D = 6$ where the round hole had a low velocity for $y/D < 0.5$, and a strong velocity gradient between $0.5 < y/D < 1.0$, and the shaped hole has higher velocities and a gradual velocity gradient. The larger region of decelerated flow downstream of the round hole may be attributed to a larger blockage by the exiting coolant jet because of the higher trajectory coming from the round hole.

A comparison of turbulence levels above and downstream of the round and shaped holes is presented in Figure 8. Turbulence levels above the holes were similar for both hole geometries. A distinct difference was found in the turbulence levels downstream of the holes. At $x/D = 6$ the peak turbulence level for the round hole was $TL > 14\%$, whereas it was $TL > 10\%$ for the shaped hole. The 40% higher turbulence level for the round hole is likely due to the strong mean velocity gradient at that location as described previously.

CONCLUSIONS

The shaped holes showed an improved performance over the round hole for all the momentum flux ratios considered in this study. For momentum flux ratios low enough so that the round hole coolant jets remained attached, the shaped holes had a marginal better performance. However, for increasing momentum flux ratios, coolant jets from the shaped holes remained attached resulting in significantly improved performance. Compared at optimum performance blowing ratios, the shaped holes had approximately 30% greater spatially averaged adiabatic effectiveness. Furthermore, the shaped holes had a much greater range of momentum flux ratios for which good adiabatic effectiveness levels were maintained.

The reason for improved performance of the shaped holes was evident from the thermal field and velocity field results which showed that shaping the round hole greatly improved the flow of the coolant jet along the surface. Consequently the coolant jets remained attached to the surface downstream of the hole even at high momentum flux ratios. Another advantage of shaping the holes was the improved lateral coverage resulting in the merging of coolant jets fairly close to the hole exit. The larger forward expansion results show that the proper design of the hole shape is critical in optimizing performance. Although the intent of the increased forward expansion angle was to maintain an attached coolant jet at even higher momentum flux ratios, performance was degraded because separation occurred within the hole.

In summary, when forced to use holes with large injection angles, shaping of the coolant hole exits was found to more than offset the detrimental effects of the large angle injection. Ultimately performance with shaped holes with large angle injection was superior to round holes with the standard 35° angle injection.

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